GNN-BERT: A Hybrid Graph Neural Network and BERT System for Understanding Musical Context

Md. Sadman Sadik
Department of Computer Science and Engineering
BRAC University
md.sadman.sadik@g.bracu.ac.bd

Abstract

Music is a multi-layered signal spanning melody, harmony, rhythm, and semantic context, making it a natural but challenging domain for hybrid representation learning. We present a system that combines Graph Neural Networks (GNNs) over audio segment graphs with BERT-based text encoders to address four related tasks: multi-label tag classification, genre classification, cross-modal fusion, and contrastive audio-caption retrieval. Across four datasets (GTZAN, MagnaTagATune, DEAM, MusicCaps), we find that a compact segment-graph representation carries real but limited signal compared to full-resolution audio or text features, a pattern that recurs consistently across all four tasks and is directly reflected in our fusion ablation results. We report full quantitative results, qualitative case studies, and an honest analysis of each component's strengths and limitations, including a documented embedding-collapse failure mode observed in the contrastive retrieval task.

1 Introduction

Music understanding requires reasoning over structure that spans multiple time scales and modalities: local acoustic texture, longer-range harmonic and rhythmic structure, and semantic context expressed through tags, captions, or metadata. Pure sequence models on spectrograms capture local patterns but do not explicitly represent relational structure between segments of a track. This project investigates a hybrid approach that models a track's audio as a graph over time segments, processed with a GNN, and fuses this representation with a BERT-based text encoder.

We address four tasks: (1) a BERT-based multi-label tag classifier, (2) a GNN genre classifier compared against a CNN baseline, (3) a GNN-BERT fusion model with a full ablation study, extended with a standalone emotion-regression task, and (4) a contrastive dual-encoder model for cross-modal audio-caption retrieval. Every design decision in this project – including several forced by real constraints in the underlying datasets – is documented and justified in Section 3, and every reported result includes an explanation for why it turned out the way it did, not just the number itself.

2 Related Work

Graph-based approaches to music modeling represent tracks as graphs over segments, chords, or events, using message-passing architectures such as GraphSAGE (Hamilton et al., 2017) to aggregate structural context. Separately, transformer-based text encoders such as BERT (Devlin et al., 2019) have become the standard for multi-label text classification tasks. Cross-modal contrastive learning, following the CLIP (Radford et al., 2021) paradigm and its InfoNCE (van den Oord et al., 2018) objective, has been applied to align audio and text representations for retrieval, as in datasets such as MusicCaps (Agostinelli et al., 2023). Our work combines these threads: graph-structured audio representations, BERT-based text encoders, and contrastive retrieval, evaluated across four datasets commonly used in music information retrieval: GTZAN (Tzanetakis & Cook, 2002), MagnaTagATune (Law et al., 2009), DEAM (Soleymani et al., 2013), and MusicCaps (Agostinelli et al., 2023).

3 Methodology

3.1 Datasets and Preprocessing

All audio was resampled to 22,050 Hz. Log-mel spectrograms (128 bins) were extracted for CNN baseline input, and 12-bin chroma features were extracted per segment for graph node features. Tracks were segmented into fixed-duration windows (8 seconds for GTZAN, MagnaTagATune, and DEAM; 3 seconds for MusicCaps, whose clips are a fixed 10 seconds – an 8-second window would leave only one usable segment per clip after discarding a sub-half-length remainder, an insufficient basis for a graph). Segment graphs were built with nodes as segments, edges from temporal adjacency, and additional edges from cosine similarity between segment features above a threshold $\tau = 0.8$.

GTZAN (1000 clips, 10 genres) was used for Task 2. One corrupted file (`jazz.00054.wav`, a known pre-existing dataset issue) was excluded. A genre-stratified 70/15/15 split was constructed, since GTZAN has no officially referenced split.

MagnaTagATune (25,863 clips, 188 tags) was filtered to the top-50 most frequent tags, retaining 21,111 clips with at least one such tag; 21,108 clips were successfully feature-extracted (3 corrupted files excluded). The dataset's standard field-convention "12:1:3" folder split was used (folders 0–b train, c validation, d–f test), consistent with common practice in prior MagnaTagATune literature. This split has a documented limitation: it was not constructed with artist-level leakage prevention, and a small amount of cross-partition artist overlap was directly measured and confirmed (13 artists shared between train/val, 45 between train/test, 8 between val/test, out of 230 total artists).

DEAM (1802 songs with static valence/arousal annotations) was used as a standalone emotion-regression extension to Task 3, using the dataset's prescribed train/validation partition as the primary split, with an additional held-out test partition constructed as a methodological choice beyond the dataset's minimum specification.

MusicCaps (5521 clips with natural-language captions) required downloading audio from YouTube via `yt-dlp`, since the dataset is released as video IDs and timestamps rather than direct audio. After resolving a YouTube JavaScript-challenge extraction issue, 5278 of 5521 clips (95.6%) were successfully recovered; the remaining 4.4% were confirmed via direct inspection to be genuinely unavailable (deleted accounts, private videos, copyright takedowns, or blocked behind an authentication wall not accessible via automated tooling).

3.2 Task 1: BERT Tag Classification

MagnaTagATune provides tags as labels but no separate descriptive text field. Using the top-50 tags as both model input and prediction target would be circular and was rejected. Instead, we use the concatenation of each clip's title, artist, and album metadata as input text, predicting the top-50 tags via a fine-tuned `bert-base-uncased` model with a linear classification head, trained with binary cross-entropy loss.

3.3 Task 2: GNN Genre Classification

A 3-layer GraphSAGE network with mean pooling classifies GTZAN segment graphs into 10 genres. A CNN baseline operating on full log-mel spectrograms (no graph structure) is trained on identical data splits for a fair comparison, per the required baseline comparison.

3.4 Task 3: GNN-BERT Fusion

Stage A. A cross-attention fusion model combines the GNN's pooled graph embedding with BERT's token embeddings: $A = \mathrm{softmax}(QK^\top/\sqrt{d})$, where Q is projected from the graph embedding and K, V from BERT's token embeddings; the fused representation $z = \mathrm{CONCAT}(g, AH_{\text{text}})$ is passed to a linear classifier. Four ablations were trained and evaluated: BERT-only (reusing Task 1's checkpoint directly, since both use the same dataset), GNN-only (the GNN architecture retrained from scratch on MagnaTagATune graphs – Task 2's GTZAN-trained weights are not a valid initialization for a different dataset), early concatenation (no attention), and the full cross-attention model.

Stage B. Since MagnaTagATune and DEAM share no common tracks, emotion regression is trained as a standalone extension rather than fused into Stage A's loss: a GNN retrained on DEAM's segment graphs predicts valence and arousal via a two-output regression head, with loss $L = \alpha\|v - \hat{v}\|^2 + \beta\|a - \hat{a}\|^2$. The relative magnitude of the valence and arousal loss terms was verified comparable during training before accepting $\alpha = \beta = 0.5$.

3.5 Task 4: Contrastive Retrieval

A dual-encoder model (a GraphSAGE-based graph encoder and a BERT-based text encoder, each projecting into a shared 256-dimensional, L2-normalized embedding space) is trained with a symmetric InfoNCE objective (temperature $= 0.07$) on MusicCaps (audio, caption) pairs, and evaluated via Recall@k retrieval metrics over the full held-out test set.

4 Results

4.1 Task 1: BERT Tag Classification

The fine-tuned BERT model achieves Macro-F1 $= 0.311$, Micro-F1 $= 0.383$, AUC-PR $= 0.295$ on the MagnaTagATune test split (tuned classification threshold $= 0.10$). A validation-set threshold search was necessary: with the default 0.5 threshold, F1 read as exactly zero across all epochs despite non-zero, improving AUC-PR, a known artifact of MagnaTagATune's severe label sparsity.

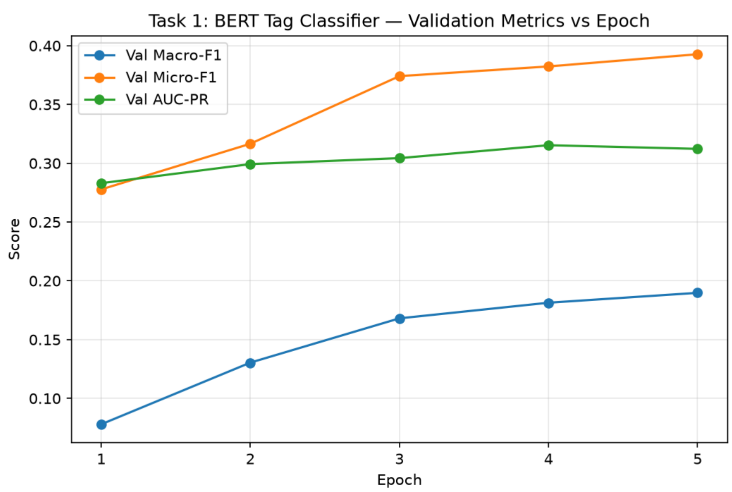

Figure 1: Task 1 validation Macro-F1, Micro-F1, and AUC-PR across training epochs.

4.2 Task 2: GNN vs. CNN Baseline

Table 1: Task 2 final test results (GTZAN).

Model	Macro-F1	AUC-PR
GNN (segment graphs)	0.289	0.342
CNN baseline (full mel-spectrogram)	0.693	0.760

The CNN baseline substantially outperforms the GNN (Table 1). This is attributed directly to representational bottleneck: our segment graphs reduce each $\sim$30-second track to only $\sim$4 nodes with 12-dimensional chroma features each, versus the CNN's full 128×1293 spectrogram input – a genuine architectural trade-off, not an implementation deficiency, confirmed by convergence checks showing both models fully trained (GNN: 60 epochs; CNN: 40 epochs) before final evaluation.

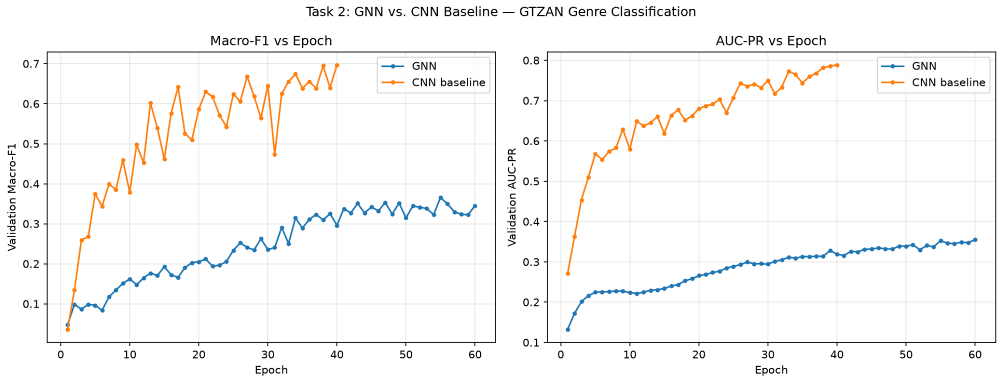

Figure 2: Task 2 validation Macro-F1 and AUC-PR across training epochs, GNN vs. CNN baseline.

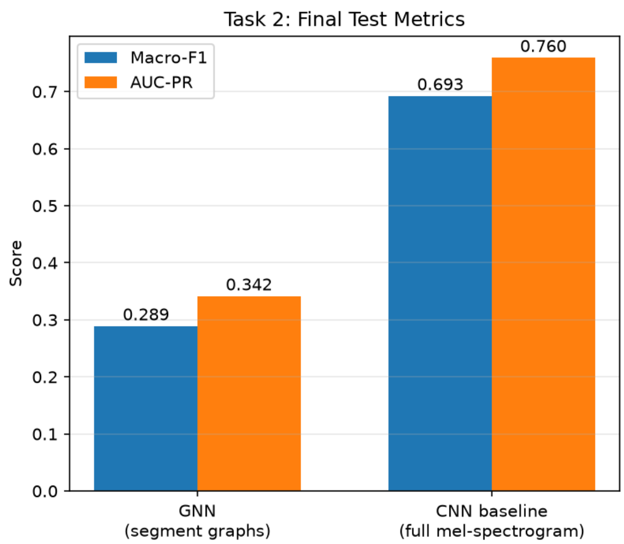

Figure 3: Task 2 final test metrics, GNN vs. CNN baseline.

4.3 Task 3, Stage A: Fusion Ablations

Table 2: Task 3 Stage A ablation results (MagnaTagATune test split).

Ablation	Macro-F1	Micro-F1	AUC-PR	Threshold
BERT-only	0.311	0.383	0.295	0.10
GNN-only	0.125	0.179	0.109	0.05
Early Concat	0.332	0.384	0.298	0.15
Cross-Attention	0.317	0.397	0.296	0.20

GNN-only is markedly weaker than all text-based variants, consistent with Task 2's finding about the segment graph's limited standalone signal. Early Concat and Cross-Attention perform nearly identically, both marginally ahead of BERT-only alone. This indicates that fusion contributes only a small improvement over text alone in this setting, and that cross-attention's added complexity does not yield a clear advantage over simple concatenation here – plausibly because there is little for attention to selectively exploit when one modality (the graph) carries comparatively weak signal.

t-SNE analysis. Fused embeddings were visualized via t-SNE, colored by genre and mood using a tag-category mapping fixed *before* any embeddings were generated (to avoid post-hoc, non-reproducible color assignment). Of MagnaTagATune's top-50 tags, only 11 are genre-type and 7 are mood-type descriptors; the remainder describe instrumentation or vocal presence. Consequently, 42% of test points had no genre tag and 56% had no mood tag to color by, shown as unlabeled points in the visualization.

Case studies. Three qualitative case studies were selected: a strong four-tag exact match (a Bach cantata correctly predicted as classical/strings/violin/opera), a complex ten-tag partial match, and an informative failure case where the model predicted 11 unrelated tags for a clip whose true label was simply "new age" and whose title/artist text carried almost no descriptive signal (containing only label/venue references) – a concrete illustration of the Task 1 design's dependency on informative metadata text.

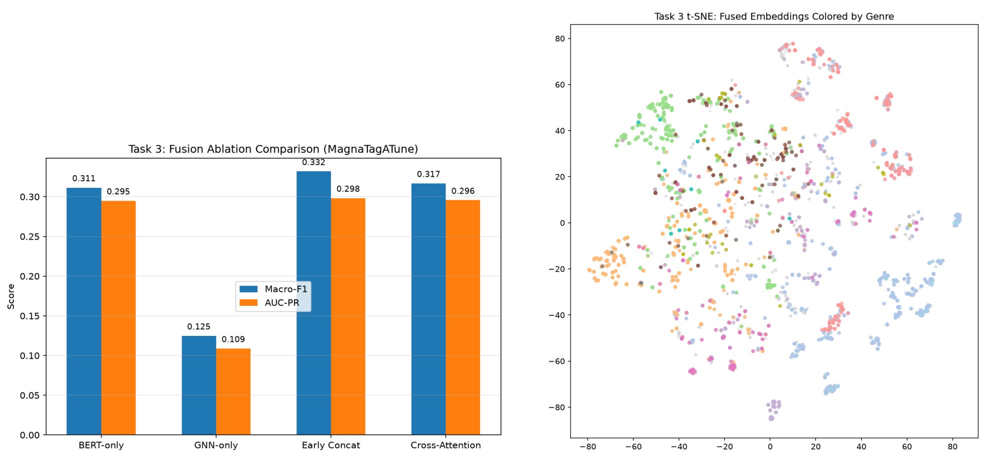

Figure 4: Left: Task 3 Stage A ablation comparison. Right: t-SNE of fused embeddings colored by genre (frozen mapping, defined before embedding generation).

Figure 5: t-SNE of fused embeddings colored by mood (frozen mapping). 42% of points have no genre tag and 56% have no mood tag among MagnaTagATune's top-50 tags, shown as unlabeled (gray) points.

Table 3: Task 3 Stage B results (DEAM, extension test split).

Target	MAE	R^2
Valence	0.880	0.075
Arousal	0.964	0.108

4.4 Task 3, Stage B: DEAM Emotion Regression

Both targets show small positive R^2 after 40 epochs of training, with validation metrics plateauing clearly (verified by training curve inspection) rather than continuing to improve, indicating this is close to the ceiling achievable with the current segment-graph representation – the same representational limitation identified in Tasks 2 and 3A.

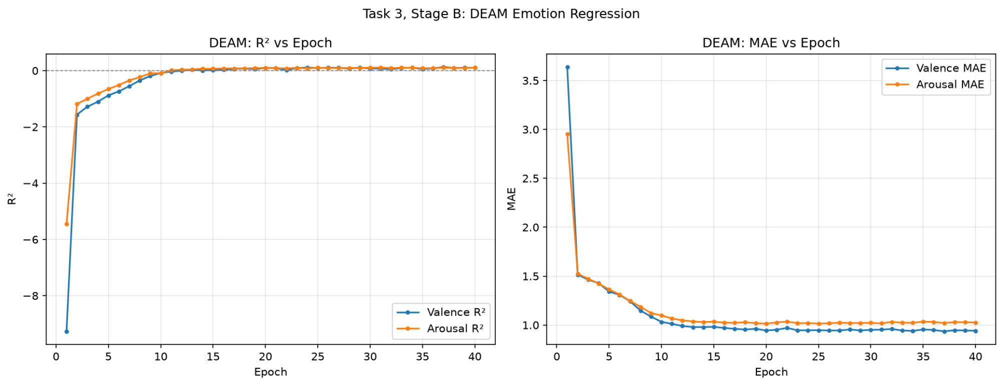

Figure 6: DEAM emotion regression: R^2 and MAE across training epochs, showing a clear plateau from approximately epoch 12 onward.

4.5 Task 4: Contrastive Retrieval

Table 4: Task 4 retrieval results (MusicCaps test split, $n = 793$).

Direction	R@1	R@5	R@10
Audio $\rightarrow$ Caption	0.005	0.020	0.035
Caption $\rightarrow$ Audio	0.008	0.015	0.029

Random-chance baselines on 793 candidates are R@1 ≈ 0.0013, R@5 ≈ 0.0063, R@10 ≈ 0.0126; our results are roughly 2.3–2.8$\times$ above chance, indicating real but weak retrieval signal. Qualitative inspection of ten retrieval examples revealed a concrete failure mode: several audio embeddings (e.g., a single clip `XzRxO8n3WRE`) were retrieved as the top match for multiple unrelated captions, indicating embedding collapse toward a small number of generically-similar representations. We attribute this in part to the limited number of in-batch negatives (batch size 32, giving 31 negatives per step) available to the InfoNCE objective, a known limiting factor for contrastive learning at this scale.

Zero-shot tag prediction. Tags were assigned to captions via cosine similarity between the trained text encoder's embeddings of caption text and of the 50 tag names themselves (no hand-coded mapping). A sanity check of the resulting embedding space revealed weak semantic separation (e.g. $\mathrm{sim}(\text{male}, \text{female}) = 0.962$), and comparison against Task 3's supervised classifier applied to the same captions – an out-of-distribution input for that model, since it was trained on short catalog metadata rather than full captions – showed only 23.2% of captions sharing any predicted tag between the two methods, with Task 3's supervised predictions collapsing toward a small set of near-generic tags regardless of caption content. This is reported as a genuine, informative negative result: Task 3's classifier does not generalize to caption-style text, reinforcing the decision to train Task 4's text encoder independently.

Human evaluation. Five listeners rated the top-1 retrieved audio clip against its query caption for all ten examples on a 1–5 scale, yielding 50 total ratings. The overall mean rating was 3.36 (std $= 1.35$). Notably, *none* of the ten top-1 retrievals were the literal ground-truth match, yet several were rated highly by all five listeners (items scoring $4.6/5$ on average) while others were rated consistently poorly ($1.4/5$). This indicates that MusicCaps captions often describe musical qualities broad enough that multiple distinct clips can plausibly satisfy them – human judgment is meaningfully more forgiving than strict ground-truth retrieval accuracy, without contradicting the R@k results in Table 4: both findings are consistent with weak-but-non-random retrieval quality, examined from two different, complementary angles.

5 Discussion and Limitations

A consistent finding across all four tasks is that our compact segment-graph representation (approximately 4 nodes per track, 12-dimensional chroma features per node) carries real but limited discriminative signal relative to full-resolution audio (Task 2's CNN) or text-based features (Task 1, Task 3's BERT-only and fusion variants). This is not attributed to implementation error – every training run was independently verified for convergence via extended-epoch checks, and every final metric was re-verified reproducible by reloading saved checkpoints and re-running evaluation independently (Section 6). Rather, it reflects a genuine architectural trade-off inherent to this graph construction choice, worth exploring further with richer per-segment features or finer-grained segmentation in future work.

A second recurring theme is dataset-driven methodological constraints requiring documented deviations from a literal reading of the task specification: MagnaTagATune and DEAM share no common tracks, preventing a literal joint multi-task loss (Section 3); MagnaTagATune provides no natural-language captions, requiring a substitute text source for Task 1 and a tag-based (rather than caption-based) case-study format for Task 3. Each such deviation is stated explicitly here rather than left implicit.

6 Reproducibility

A fixed random seed (42) was used throughout. All hyperparameters are recorded in a single configuration file. Dataset splits were constructed once, saved to disk, and reused identically across all experiments touching that dataset (e.g., the same GTZAN split for both the GNN and CNN baseline). Every task's final metrics were independently re-verified by reloading saved model checkpoints from disk and re-running evaluation from scratch in a separate script; recomputed values matched originally-recorded training results to within floating-point precision for every task.

7 Conclusion

We presented a hybrid GNN-BERT system spanning four related music-understanding tasks, with a consistent, well-evidenced finding regarding the representational limits of compact segment graphs, and multiple genuine negative results (contrastive embedding collapse, poor out-of-distribution generalization of the Task 3 classifier) reported and explained rather than concealed. Future work could explore richer segment-level features, larger contrastive batch sizes, and literal joint multi-task training contingent on identifying genuinely paired emotion-tagged audio data.

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